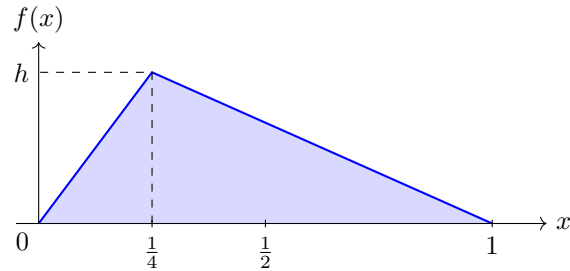


Math 372: Homework 07

Due Wednesday, October 14

Problem 1. A continuous random variable X has the density curve shown below:



(a) Use the area under the curve to determine h .

(b) What is $\mathbb{P}(X = \frac{1}{4})$? Explain how your answer is consistent with the height of the density at this point.

(c) Determine which event is more likely by computing their probabilities using geometry:

$$\left[0 < X < \frac{1}{4}\right] \quad \text{or} \quad \left[\frac{1}{4} < X < \frac{1}{2}\right].$$

Problem 2. Suppose X has pdf

$$f(x) = \begin{cases} 2 & : 0 < x < \frac{1}{2} \\ 0 & : \text{otherwise} \end{cases}$$

- (a) Plot the density curve $y = f(x)$.
- (b) A student says this is invalid because probabilities cannot exceed 1. Explain their mistake.
- (c) What is $\mathbb{P}(X = \frac{1}{4})$?
- (d) What is $\mathbb{P}(0 < X < \frac{1}{4})$?
- (e) Find the cdf F of X . Plot it.
- (f) Why is the cdf increasing on the interval $(0, \frac{1}{2})$ even though the pdf is constant?

Definition 3. If X is a continuous random variable with pdf f , then the *expected value* of X is

$$\mathbb{E}[X] := \int_{-\infty}^{\infty} xf(x)dx,$$

provided that this integral converges absolutely.

Problem 4. Suppose X is a random variable with pdf

$$f(x) = \begin{cases} 3x^2 & : 0 \leq x \leq 1 \\ 0 & : \text{otherwise} \end{cases}$$

(a) Sketch the density curve.

(b) Verify that f is a valid pdf (i.e., show that f is nonnegative and that the area under the curve is 1)

(c) Compute $\mathbb{P}[0 < X < \frac{1}{2}]$ and $\mathbb{P}[\frac{1}{2} < X < 1]$. Explain why these intervals have different probabilities even though they have the same width.

(d) Compute the mean $\mathbb{E}[X]$. Is X more likely to be larger or smaller than its mean? Justify your answer with a calculation.

Problem 5. Suppose you station yourself on a spot on an idyllic country road, using a stopwatch to count the time between passing cars. Let X be the time in minutes between the passing of one car and the arrival of the next. Suppose X has pdf

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & : x > 0 \\ 0 & : x \leq 0, \end{cases}$$

where $\lambda > 0$ is a fixed parameter. (**Note:** a random variable with this pdf is called an *exponential* random variable with rate λ , and we write $X \sim \exp(\lambda)$.)

(a) Find $\mathbb{P}[X > x]$ for all $x \in \mathbb{R}$, and use this to find the cdf F of X .

(b) Now suppose $\lambda = 2$. The average wait time (in minutes) between cars is $\mathbb{E}[X]$. Compute this quantity [hint: *integration by parts*] and use your answer to interpret the parameter λ .

(c) Suppose $\lambda = 2$. Is the next car more likely to arrive before or after the average waiting time? Do a calculation and explain.

(d) Suppose $\lambda = 2$. What is the probability that you must wait at least 30 seconds for the next car?

(e) Suppose $\lambda = 2$. Given that one minute has already passed since the last car, and no new car has arrived yet, what is the probability that you must wait at least 30 more seconds. Compare your answer to (d).

Problem 6. For a continuous random variable, the probabilities

$$\mathbb{P}[a < X < b], \quad \mathbb{P}[a \leq X < b], \quad \text{and} \quad \mathbb{P}[a \leq X \leq b]$$

are all equal. Explain why.